

Deligne's Letter to Vasiu

November 30, 2011

Dear Vasiu,

Let $\mathcal{T}$ be a tannakian category over an algebraically closed field k . Your question amounts to: does $\mathcal{T}$ always have a fiber functor over k ? I believe it is true. The following argument is presumably too pedestrian, hiding which “compactness arguments” are really relevant.

Let $\mathcal{A}$ be the set of strictly full subcategories of $\mathcal{T}$, stable by $\otimes$, subquotients and duals. Let $\mathcal{A}_f \subset \mathcal{A}$ be the set of those generated by finitely many objects, using the same operations. For $\alpha \in \mathcal{A}$, I denote by $\mathcal{T}_\alpha$ the corresponding subcategory (as defined: $\mathcal{T}_\alpha = \alpha$). The set $\mathcal{A}$ is ordered by inclusion. I assume we already know the existence and unicity up to isomorphism of fiber functors for the $\mathcal{T}_\alpha$, $\alpha \in \mathcal{A}_f$.

If $\mathcal{T}_\alpha \subset \mathcal{T}_\beta$, it makes sense to say that a fiber functor ω_β on $\mathcal{T}_\beta$ extends (on the nose) a fiber functor ω_α on $\mathcal{T}_\alpha$. If an extension up to isomorphism exists, an actual extension exists too.

Let us order the set of pairs

$$(\alpha, \omega_\alpha), \quad \alpha \in \mathcal{A}, \quad \omega_\alpha \text{ a fiber functor on } \mathcal{T}_\alpha,$$

by “ ω_β extends ω_α ”. To avoid set-theoretical difficulties, one could consider only the fiber functors ω_α such that the $\omega_\alpha(X)$ take values in the set of vector spaces k^n , $n \in \mathbb{N}$.

The ordered set of (α, ω_α) is inductive (Bourbaki, *Ens.*, Ch. 3, §2, no. 4): if I is a totally ordered subset, the “union” of the (α, ω_α) is a $\mathcal{T}_0$ in $\mathcal{A}$,

$$\mathcal{T}_0 = \bigcup_{(\alpha, \omega_\alpha) \in I} \mathcal{T}_\alpha,$$

with the fiber functor ω_0 characterized by $\omega_0|_{\mathcal{T}_\alpha} = \omega_\alpha$. This “union” majorizes I . By Bourbaki, loc. cit., Th. 2, the ordered set of the (α, ω_α) has a maximal element $(\mathcal{T}_1, \omega_1)$. To prove that $\mathcal{T}_1 = \mathcal{T}$, it suffices to prove the following.

Lemma 1. Let $\mathcal{T}'$ be in $\mathcal{A}$ and let $\mathcal{T}''$ be in $\mathcal{A}_f$. Let $\langle \mathcal{T}', \mathcal{T}'' \rangle$ be the category generated by $\mathcal{T}'$ and $\mathcal{T}''$; it is in $\mathcal{A}$. Then any fiber functor ω' on $\mathcal{T}'$ can be extended (on the nose, or up to isomorphism; this amounts to the same) to $\langle \mathcal{T}', \mathcal{T}'' \rangle$.

I first prove:

Lemma 2. Suppose $\mathcal{T}'$ is in $\mathcal{A}_f$ too. “Restriction” is then an equivalence of categories

$$\left\{ \text{fiber functors } \omega \text{ on } \langle \mathcal{T}', \mathcal{T}'' \rangle \right\} \xrightarrow{\sim} \left\{ \begin{array}{l} \text{triples of a fiber functor } \omega' \text{ on } \mathcal{T}', \omega'' \text{ on } \mathcal{T}'', \\ \text{and an isomorphism } \tau \text{ of the restrictions of } \omega' \text{ and } \omega'' \\ \text{to } \mathcal{T}' \cap \mathcal{T}'' \text{ (} \mathcal{T}' \cap \mathcal{T}'' \text{ is also in } \mathcal{A}_f \text{)} \end{array} \right\}.$$

We may assume that $\langle \mathcal{T}', \mathcal{T}'' \rangle$ is the category of representations of a group G , and that, for invariant subgroups A and B , $\mathcal{T}'$ (resp. $\mathcal{T}''$) is the subcategory of representations where A (resp. B) acts trivially. That they generate $\langle \mathcal{T}', \mathcal{T}'' \rangle$ means that $A \cap B = \{e\}$. The intersection $\mathcal{T}' \cap \mathcal{T}''$ is the category of representations on which AB acts trivially.

The triples $(\omega', \omega'', \tau)$ are all isomorphic: as all ω' (resp. all ω'') are isomorphic, it suffices to see that $(\omega', \omega'', \tau_1)$ and $(\omega', \omega'', \tau_2)$ are isomorphic. Indeed, τ_1 and τ_2 differ by an automorphism of $\omega'|_{\mathcal{T}' \cap \mathcal{T}''}$, and such an automorphism lifts to an automorphism of ω' :

$$G/A(k) \longrightarrow G/AB(k) \quad \text{is onto.}$$

We hence have here categories with just one isomorphism class of objects, and the question is to compare automorphism groups. We need to check

$$G \xrightarrow{\sim} \{(g', g'') \in G/A \times G/B \mid g' \text{ and } g'' \text{ have the same image in } G/AB\}.$$

Equivalently, this is the Cartesian square

$$\begin{array}{ccc} G & \longrightarrow & G/A \\ \downarrow & & \downarrow \\ G/B & \longrightarrow & G/AB. \end{array}$$

Injectivity of this morphism of groups amounts to $A \cap B = \{e\}$. For surjectivity, if $\tilde{g}', \tilde{g}''$ lift g', g'' , then, after writing $\tilde{g}'' = \tilde{g}'ab^{-1}$ with $a \in A, b \in B$, the element $\tilde{g}'a = \tilde{g}''b \in G$ maps to (g', g'') .

Lemma 3. The same as Lemma 2, but with $\mathcal{T}'$ only assumed to be in $\mathcal{A}$.

Let $\mathcal{B}$ be the set of the $\mathcal{T}_\alpha$ ($\alpha \in \mathcal{A}_f$) contained in $\mathcal{T}'$. One has

$$\langle \mathcal{T}', \mathcal{T}'' \rangle = \bigcup_{\beta \in \mathcal{B}} \langle \mathcal{T}_\beta, \mathcal{T}'' \rangle.$$

One also has equivalences

$$\{\text{fiber functors on } \mathcal{T}'\} \xrightarrow{\sim} \left\{ \begin{array}{l} \text{fiber functors } \omega_\beta \text{ on the } \mathcal{T}_\beta, \text{ plus a compatible system} \\ \text{of isomorphisms } \omega_\beta|_{\mathcal{T}_\gamma} \xrightarrow{\sim} \omega_\gamma \text{ for } \mathcal{T}_\gamma \subset \mathcal{T}_\beta, \\ \text{with the compatibility condition for } \mathcal{T}_\delta \subset \mathcal{T}_\gamma \subset \mathcal{T}_\beta \end{array} \right\}.$$

The same holds for fiber functors on $\langle \mathcal{T}', \mathcal{T}'' \rangle = \bigcup \langle \mathcal{T}_\beta, \mathcal{T}'' \rangle$.

The $\mathcal{T}_\beta \cap \mathcal{T}''$ have a largest element: if $\mathcal{T}'' = \text{Rep}(G'')$, they correspond to invariant subgroups of G'' ; subgroups are closed subschemes, and one uses the noetherian property. If β_0 is such that $\mathcal{T}_{\beta_0} \cap \mathcal{T}''$ is the largest $\mathcal{T}_\beta \cap \mathcal{T}''$, then for any $\beta \geq \beta_0$, extending $\omega_\beta = \omega|_{\mathcal{T}_\beta}$ to $\langle \mathcal{T}_\beta, \mathcal{T}'' \rangle$ amounts to extending $\omega_\beta|_{\mathcal{T}_{\beta_0} \cap \mathcal{T}''}$ to $\mathcal{T}''$ (Lemma 2), that is, to extending ω_{β_0} from $\mathcal{T}_{\beta_0} \cap \mathcal{T}''$ to $\mathcal{T}''$. If we choose one such extension, we get, up to unique isomorphism, a system of extensions of the ω_β to $\langle \mathcal{T}_\beta, \mathcal{T}'' \rangle$, and by gluing them an extension of ω to $\langle \mathcal{T}', \mathcal{T}'' \rangle$.

This is all we need to conclude that $\mathcal{T}' = \mathcal{T}$.

The proof should be cleaned up. After all, we are proving that some projective system of gerbes (of the fiber functors on the $\mathcal{T}_\alpha$, $\alpha \in \mathcal{A}_f$), where “projective system” is taken in a 2-categorical sense, has a nonempty projective limit (again a limit in a 2-categorical sense). Maybe such a translation would make the “compactness arguments” used clearer. There were two of them: “fiber functor is a property of finite type” (cf. Bourbaki, *Ens.*, Ch. 3, §4, no. 5), and the noetherian property for subgroups.

The same arguments give unicity up to isomorphism of ω , over k : we get a maximal $\mathcal{T}_\alpha$ ($\alpha \in \mathcal{A}$) over which we have an isomorphism, and extend further if $\mathcal{T}_\alpha \neq \mathcal{T}$.

Best,

P. Deligne